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Aparnabindu: A Hybrid ConvNeXt-Transformer Framework for Global-Local Structural Analysis of Kolam Patterns

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Abstract

Kolam is a traditional South Indian art form known for its intricate geometric patterns and symmetry, making automated classification challenging. Existing CNN-based methods mainly focus on local features and struggle to capture complex global relationships. Recent models such as KolamNet and KolamNetV2 achieve around 81% accuracy but still have limitations in modeling long-range spatial dependencies.

To address this, we propose Aparnabindu, a hybrid deep learning framework that combines ConvNeXt for local feature extraction with a Vision Transformer (ViT) for global context modeling. ConvNeXt captures fine-grained geometric patterns, whereas the ViT learns long-range relationships across the entire Kolam. Our model was evaluated on a dataset of over 12,000 augmented images across 12 classes and achieved an overall classification accuracy of 86.16%, outperforming the existing baselines. The learned feature representation also retains useful structural information that may support future Kolam generation techniques.

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1. Introduction

Kolam pattern classification is a challenging computer vision problem as recognition depends on both fine-grained details and overall geometry. Individual dots, curves, and motifs provide local information, while symmetry determines how these elements form a complete Kolam. Previous approaches such as KolamNet and KolamNetV2, have improved classification performance but still face limitations in modelling relationships between distant regions of a pattern. This becomes particularly relevant when different Kolam classes contain similar local motifs but differ in their overall arrangement.

Nomenclature

MHSA	Multi-Head Self-Attention
MLP	Multi Layer Perceptron
GELU	Gaussian Error Linear Unit
F	ConvNeXt feature map
H', W'	spatial dimensions of the feature map
D_{cnn}	CNN feature dimension
N	number of tokens
D_{vit}	Transformer embedding dimension
Q, K, V	query, key, and value matrices
d_k	key-vector dimension
w_i	class weight

1.1. Kolam Classification and Computational Approaches

Early computational approaches to Kolam classification primarily relied on handcrafted feature extraction techniques combined with classical machine learning classifiers which focused on basic edge-features and symmetry inherent in Kolam designs. While such approaches demonstrated initial feasibility, their performance was limited by noise sensitivity, variations in drawing style, as well as camera angles.

Models such as KolamNet leverage Convolutional architectures to automatically learn hierarchical visual features from Kolam images, achieving classification accuracies of approximately 81% [5]; KolamNetV2 improved upon this by introducing attention mechanisms to EfficientNet architectures [4], but still limited by localized convolutional processing.

1.2. Limitations of CNN-Based Classification

Despite their success, pure CNN-based models exhibit inherent limitations when applied to highly intricate Kolam designs. CNNs operate on local recetive fields, progressively aggregating spatial information. While effective for local patterns, it restricts the model's ability to explicitly learn long-range dependencies and global structural relationships between distant regions of an image. As the number of Kolam classes increases, CNN-only architectures often require deeper networks, leading to increased computational cost and diminishing performance gains.

1.3. Vision Transformers and Hybrid Approaches

Vision Transformers (ViT) [6] have emerged as a powerful alternative by modeling images as sequences of fixed-size patches. Through self-attention mechanisms, ViTs explicitly learn relationships between all patches, establishing global dependencies. However, standalone ViT models often demand large datasets and substantial computational resources.

Recent trends in image analysis have shown that combining both CNN and ViT into a hybrid architecture provides the best of both worlds, balancing local feature extraction with global context modeling [10]. Similar hybrid structures have successfully been applied to dense settlement detection in remote sensing [11] and architectural segmentation [12]. Taking heed from this, our architecture consists of a ConvNeXt backbone for local motif extraction as well as a ViT for global structural understanding.

1.4. Mathematical Modeling of Kolam Patterns

Prior to deep learning, Kolam patterns were studied in relation to formal language theory and discrete mathematics. Siromony et al. (1974) introduced array grammars to generate and classify Kolams [7], establishing that the designs follow strict generative rules. Later works extended on this using Psystems (also known as a membrane computer) and string rewriting rules [8], [9]. Yanagisawa and Nagata (2007) conducted a fundamental study on the design system of Kolam patterns, decomposing the art form into core syntactical elements and basic motifs to mathematically map out the generation process [13].

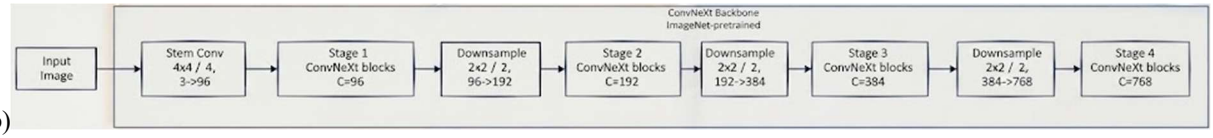
These foundational studies show that Kolam is a highly correlated geometric array whose properties justify our architectural choice, where the CNN is the syntactic analyzer, identifying curves and dots as defined in traditional design systems, while the ViT acts as the semantic analyzer, understanding the overall geometry.

2. Proposed System

(a)



(b)



(c)

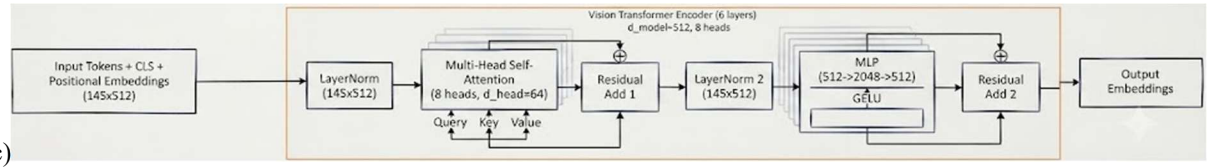


Fig. 1: Proposed Aparnabindu Architecture: (a) The overall hybrid workflow. (b) The ConvNeXt backbone for local feature extraction. (c) Vision Transformer module for global structural context modeling.

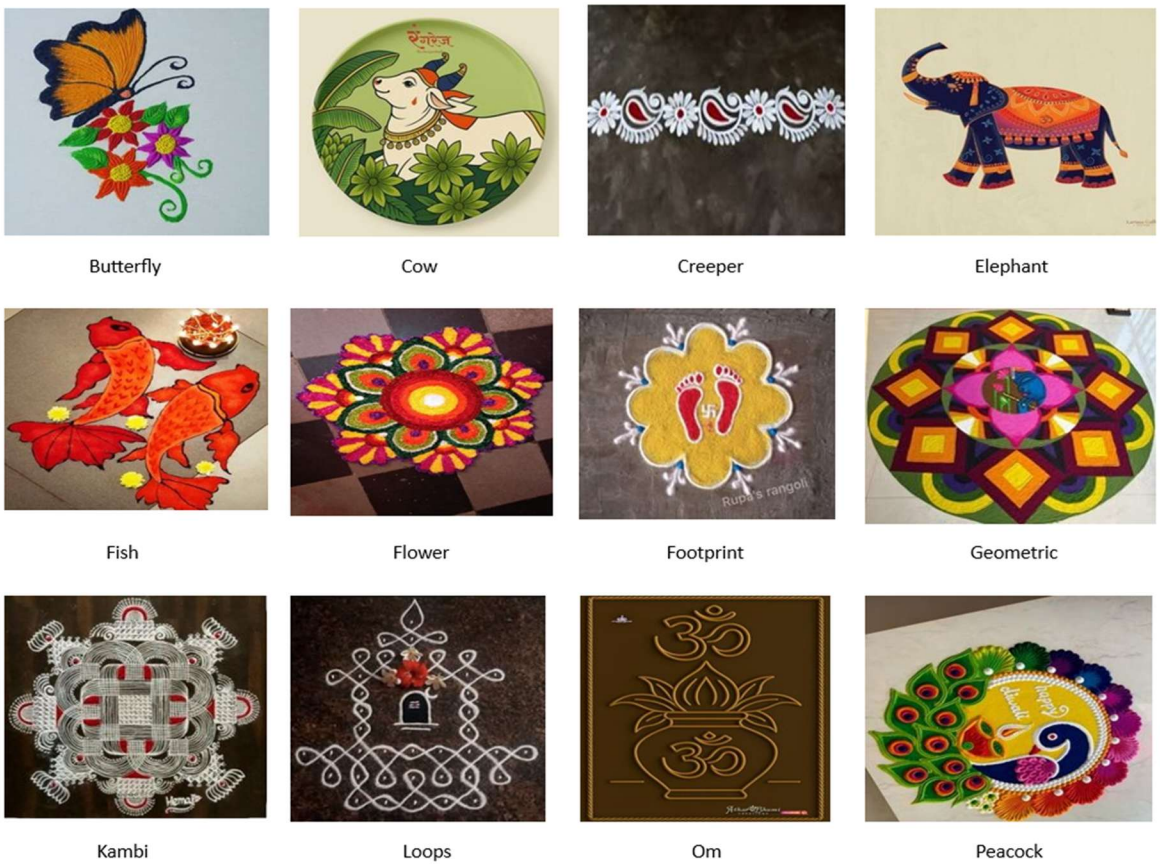


Fig. 2: Representative samples from the augmented dataset showcasing all 12 Kolam classes. The preprocessed 380 × 380 images retain the intricate structural variations required for accurate classification.

2.1. Dataset Acquisition and Preprocessing

Images were taken during Indian festivities and collected from online platforms. The initial collection of approximately 3,000 images was augmented to over 12,000 images to prevent overfitting. Specifically, transformations such as rotations ([0,360] degrees), horizontal/vertical flips, and minor perspective skews were applied with colour shifts and filters, as Kolam designs drawn on the floor are viewed from different angles.

Images were resized to 380 × 380 pixels and normalized. The dataset is organized into 12 classes: butterfly, fish, kambi, cow, flower, loops, creeper, footprint, om, elephant, geometric, peacock. Figure 2 presents samples from all 12 classes.

2.2. Hybrid Model CNN-ViT Architecture

A detailed layer-wise breakdown of the architecture, mapping the input shapes to the final classifications, is provided in Table I

Table I: Detailed Architectural Operations of the Hybrid CNN-ViT model

Stage / Layer	**Input Shape**	**Operation (Kernel Size, Stride, Channel transformation)**	**Output Shpae**
Input	380 × 380 × 3	-	380 × 380 × 3
Padding + Stem Conv	384 × 384 × 3	4×4, 4, 3 → 96	96 × 96 × 96
Stage 1 blocks	96 × 96 × 96	ConvNeXt blocks (C = 96)	96 × 96 × 96
Stage 1 Downsample	96 × 96 × 96	2×2, 2, 96 → 192	48 × 48 × 192
Stage 2 Blocks	48 × 48 × 192	ConvNeXt blocks (C = 192)	48 × 48 × 192
Stage 2 Downsample	48 × 48 × 192	2×2, 2, 192 → 384	24 × 24 × 384
Stage 3 Blocks	24 × 24 × 384	ConvNeXt blocks (C = 384)	24 × 24 × 384
Stage 3 Downsample	24 × 24 × 384	2×2, 2, 384 → 768	12 × 12 × 768
Stage 4 Blocks	12 × 12 × 768	ConvNeXt blocks (C = 768)	12 × 12 × 768
Flatten	12 × 12 × 768	-	144 × 768
Linear Proj	144 × 768	, , 768 → 512	144 × 512
Add CLS + Pos Emb	144 × 512	Learnable 1D Embeddings	144 × 512
ViT Encoder (× 6)	145 × 512	MHSA + MLP	144 × 512
CLS Extract	145 × 512	-	512
FC Classifier	512	, , 512 → 12	12

2.3. Computational Complexity Analysis

The hybrid design optimizes the computational efficiency. A ViT-only model processing a 380×380 image with standard 16 × 16 patches results in N ≈ 564 tokens, making the complexity of self-attention $O(N^2\, D_{ViT})$ a limiting factor. By utilizing the ConvNeXt backbone to reduce the spatial dimensions to a 12 × 12 grid before tokenization, we reduced the sequence length to N = 144. The spatial feature extraction burden is shifted to the CNN, which has a linear complexity $O(H'W'D_{cnn}^2)$ with respect to spatial dimensions, allowing the ViT to focus exclusively on global semantics.

2.4. ConvNeXt Block Design and Mathematical Formulation

The ConvNeXt backbone processes the raw input images. For a given tensor $X \in R^{H \times W \times C}$,

- A depthwise 7 × 7 convolution captures the local spatial context, given by:

$$X_1 = DepthwiseConv(X)$$

- Layer Normalization is applied as follows:

$$X_2 = LayerNorm(X_1)$$

- An inverted bottleneck quadruples the channel dimension using pointwise (1 × 1) convolution:

$$X_3 = GELU(W_{Up}X_2)$$

- The second pointwise convolution projects the channels back down:

$$X_4 = W_{down}X_3$$

- A residual connection is added:

$$X_{out} = X + X_4$$

2.5. Mathematical Formulation of the CNN-ViT Bridge

Let $X \in R^{H \times W \times C}$ be an input image (380 × 380 × 3). The ConvNeXt backbone outputs a feature map $R^{H' \times W' \times Dcnn}$, where $H' = 12$, $W' = 12$, and $D_{cnn} = 768$

Flatten F into a sequence of patches $Z_0 \in R^{N \times Dcnn}$, where $N = H' \times W' = 144$. This sequence was projected onto $D_{ViT} = 512$. A learnable classification token Z_{CLS} is preloaded, and 1D learnable positional embeddings E_{pos} are added to retain spatial awareness

$$Z = [Z_{CLS}\,; Z_0 W] + E_{pos} \in R^{(N+1) \times DViT}$$

2.6. Results and Discussion

The model was trained end-to-end using an AdamW optimizer with a cosine annealing learning rate scheduler. Loss function is a weighted cross-entropy loss to eliminate class imbalance. All of the performance metrics throughout the board were better for AparnaBindu compared to its predecessors.

Table II: Performance Comparison of Aparnabindu with State-of-the-Art Techniques

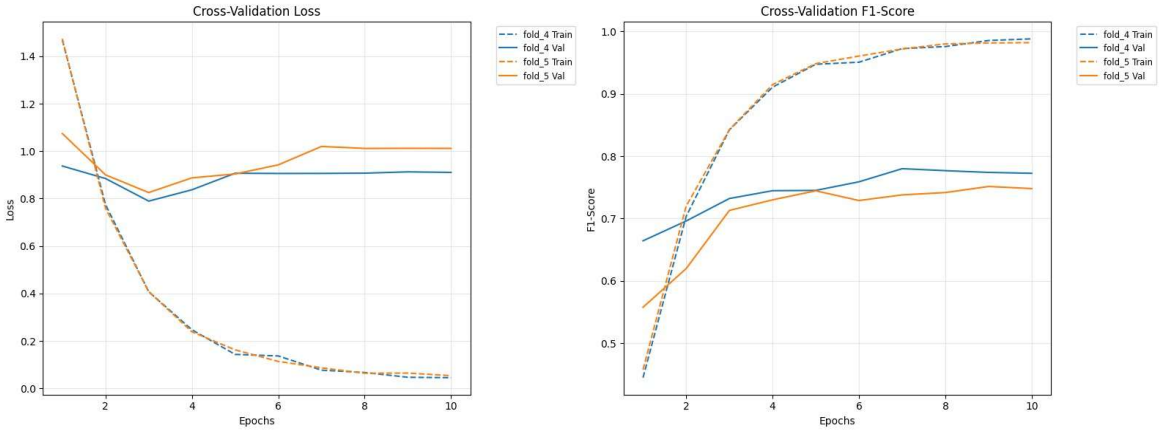
Model	Precision	Recall	F1 Score	Time (s)
Aparnabindu	**0.8740**	**0.8275**	**0.8427**	**0.03850**
Efficient B1	0.787	0.788	0.783	0.02034
Efficient B4	0.772	0.798	0.776	0.03120
KolamNet [5]	0.777	0.778	0.764	**0.01960**
KolamNetV2 [4]	0.795	0.785	0.785	0.04244

Table III: Detailed Classification Report on Test Set

Class	Precision	Recall	F1 Score	Support
butterfly	0.9375	0.8333	0.8824	18
cow	0.9149	0.9773	0.9451	44
creeper	0.8846	0.8846	0.8846	52
elephant	0.9250	0.9487	0.9367	39
fish	0.9000	0.7826	0.8372	23

flower	0.7895	0.7792	0.7843	77
footprint	0.9487	0.8409	0.8916	44
geometric	0.8605	0.8409	0.8506	44
kambi	0.7692	0.8696	0.8163	23
loops	0.9194	0.9194	0.9194	62
om	0.8571	0.7059	0.7742	17
peacock	0.8600	0.9100	0.8900	47
Accuracy	-	0.8616	-	477
Macro Avg	0.8700	0.8300	0.8400	477
Weighted Avg	0.8700	0.8600	0.8600	477

(a)



(b)

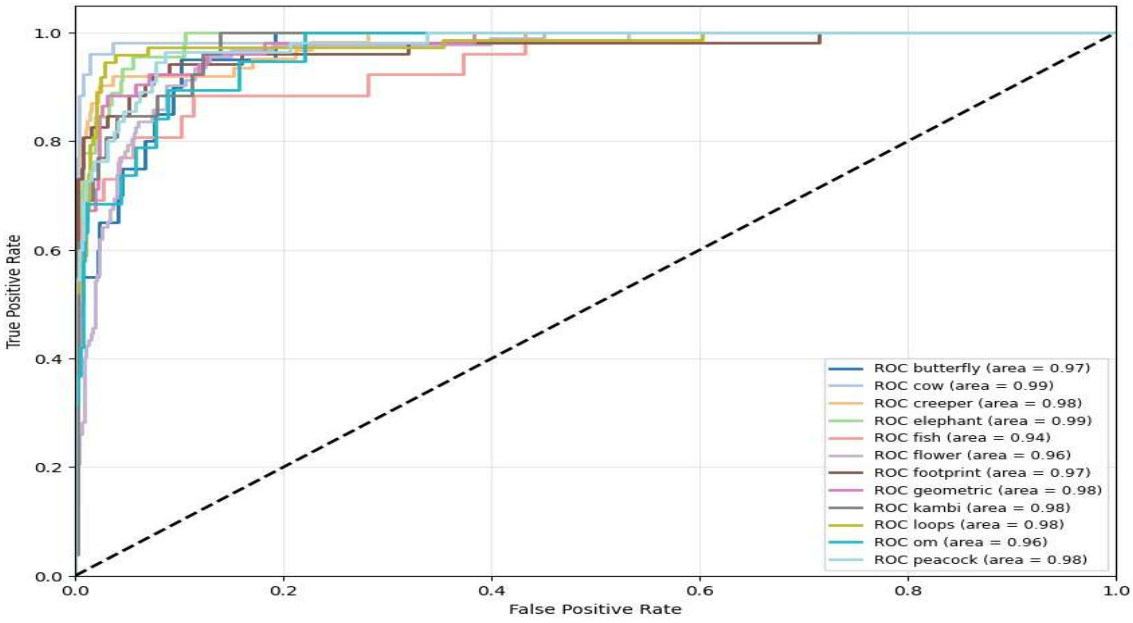


Fig. 3: Model Evaluation Metrics: (a) Cross-validation loss and F1 score progression over epochs, demonstrating stable convergence without severe overfitting. (b) Receiver Operating Characteristic (ROC) curves for the 12 Kolam classes, highlighting the high discriminative capability (AUC > 0.95) of the Aparnabindu architecture across diverse structural motifs.

B. Cross-Validation and Robustness Analysis

The dataset was divided into 5 equally sized, stratified folds, preserving the class distribution in each fold with variance in model performance across all folds being negligible, verifying the model's high generalization on structurally complex patterns. As shown in Fig. 3, the training and validation dynamics show stable convergence.

3. Ablation Studies

To justify the hybrid architecture, an ablation study isolating the local and global modeling mechanisms was tested. The performance of the isolated baselines compared to the proposed hybrid model is summarized in Table IV.

Table IV: Ablation Study: Architecture Component Analysis

Model Configuration	**Architecture Focus**	**Accuracy (%)**	**Precision**	**Recall**	**Macro F1**
Baselines					
ViT-Only (Raw Patches)	Global Attention Only	35.43	0.3124	0.3216	0.3122
ConvNeXt-Only (Avg Pool)	Local Convolution Only	80.31	0.8100	0.7700	0.7800
Aparnabindu	**Local + Global Hybrid**	**86.16**	**0.8740**	**0.8275**	**0.8427**

The standalone ViT performs poorly because it lacks the localized inductive biases inherent to CNNs when trained from scratch on a moderately sized dataset. The standalone CNN performs comparatively better, as its good at extracting local features like dots, curves, edges, and motifs. However, it hits a performance limit because it cannot connect all the features in the image to understand global symmetry. While ***Aparnabindu*** achieves higher scores across the board by combining the strengths of both versions

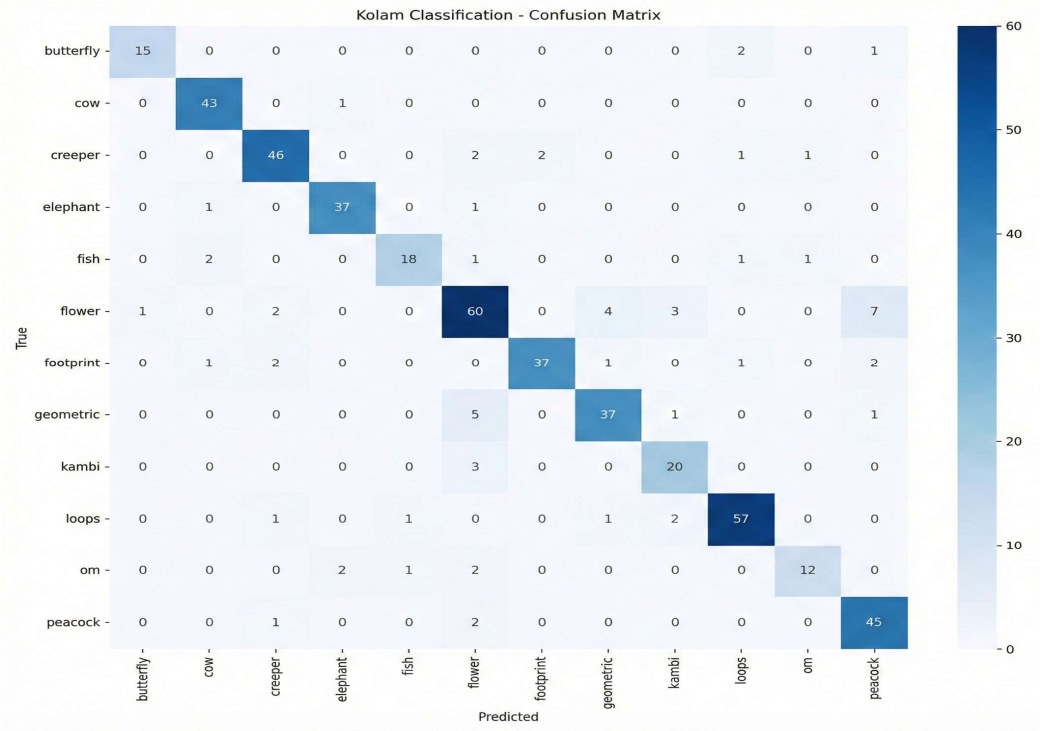


Fig. 4: Confusion Matrix Heatmap of the Proposed Aparnabindu Model

For example, there is an overlap between the ‘flower’ and ‘peacock’ classes. This is due to shared curvilinear motifs and radial symmetries present in both patterns, But the hybrid model minimizes these errors far better than traditional CNNs by using the ViT’s attention mechanism.

4. Qualitative Error Analysis

In the confusion matrix; while classes like ‘cow’ and ‘peacock’ demonstrated high precision (> 0.91), the model performed occasional misclassifications between structurally similar classes as seen in Figure 4.

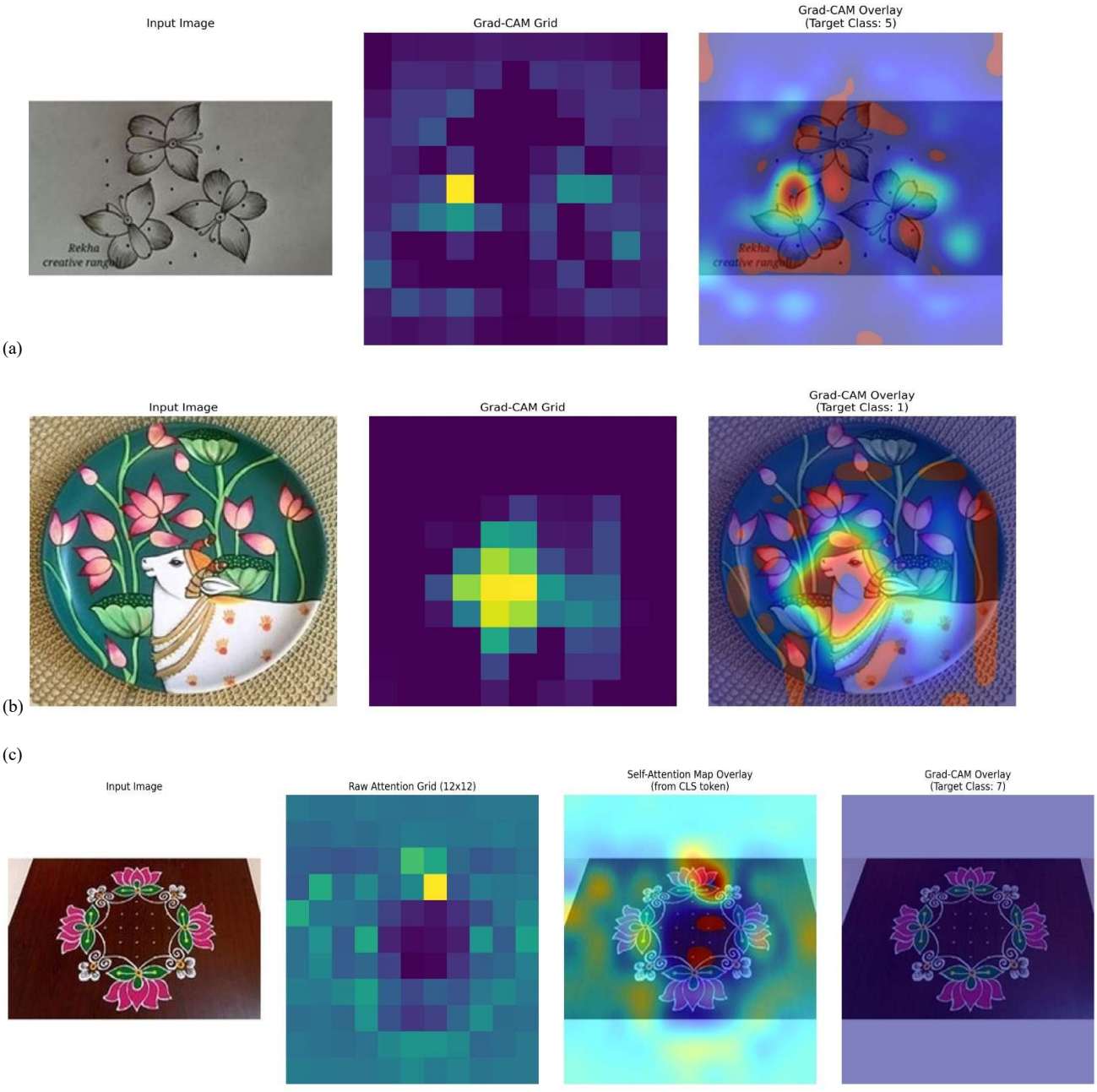


Fig. 5: (a) Butterfly Attention Map, (b) Cow Attention Map, (c) Flower Attention Map.Visualization of attention maps highlighting salient regions of Kolam motifs learned by the proposed model. The maps indicate that the network focuses on structurally significant curves and intersections of the Kolam patterns.

5. Explainability using Attention Maps

The model's decision-making behavior was evaluated using attention maps and gradientbased attribution (GradCAM), as seen in Figure 5.

Visual inspection of these maps shows that the network tends to focus on:

- Intersection points where loops meet or diverge.
- Regions around dense dot clusters in kambi Kolams.
- Axis of symmetry in motifs like peacock, flower, and loops.

These patterns align with human intuition about the structure of Kolams and suggest that the hybrid model is learning semantically meaningful representations instead of background cues.

6. Real-time Inference Evaluation

To demonstrate the practical inference capability of the hybrid CNN-ViT model, a single Kolam image was given as input and its predictions are as such in Figure 6.

Class	Prob.
Peacock	99.94%
Geometric	0.04%
Elephant	0.01%
Others	0.01%



Fig 6: Kolam with Peacock Motif and the corresponding predicted probability distribution across all classes

This experiment highlights several technical strengths:

- High Confidence Accuracy: For the test image of a complex peacock motif, the hybrid model achieved a confidence score of 99.94%.
- Real-Time Performance: The transition from ConvNeXt feature maps to ViT tokens allows for fast classification (0.038s per image), making it suitable for user-driven applications.
- Structural Robustness: The low probabilities assigned to other classes demonstrate the model's accurate discriminative power and its ability to differentiate between structurally distinct classes despite having similarities.

7. Limitations

While Aparnabindu achieves SOTA results, some limitations remain. The model occasionally struggles with heavily concealed Kolams (e.g., drawn on highly textured or uneven dirt surfaces) and multi-class patterns where one class motif overlaps another.

8. Conclusion and Future Work

This paper presented a hybrid ConvNeXt-Vision Transformer framework for Kolam pattern classification, bridging traditional Indian art with advanced computational intelligence. By integrating convolutional layers for localized geometric primitive extraction and a self-attention mechanism for global structural coherence, the proposed model

mitigates the limitations of standalone CNNs, achieving a robust accuracy of 86.16%. Future work will focus on expanding the dataset with wider regional variations, and utilizing the learned high-dimensional representations to synthesize novel Kolam patterns via Generative Adversarial Networks (GANs) and Diffusion Models.

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